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CSE 566
Project Proposal

My project follows the form of :

"*Find some recent publications within the field of HPC, and write a paper that critques them"

as mentioned in the CSE566 course webpage. I have decided to work on the paper "High-Performance Computing with Quantum Processing Units[1]".

The paper undergoes analysis of emerging Quantum Processors and their integration with High Performance Computing. The paper begins with defining Quantum Processing Units (QPU) and then addresses two integration pathways based on infrastructure constraints in processing: tight integration that assumes infrastructure bottlenecks can be overcome while loose integration that assumes that they cannot. Finally, the paper lays out challenges in assigning HPC metrics to QPU systems and considers some new metrics based on QPU architecture such as relative overhead required by fault tolerances in instruction protocols for the instruction set architecture, longest duration gates within QRAM,etc.

I would like to analyze the paper based on physical and algorithmic constraints that a QPC might have. This includes analyzing the two integration pathways suggested by the paper in terms of practicality and scalability. Finally, I would like to discuss how high performing such an architecture can become given the constraints of the architecture discussed in the paper with metrics we have discussed in class, such as parallelism and pipelining benchmarks, etc.

[1] https://arxiv.org/pdf/1511.04386.pdf